

CARISURG Triage Guardian Week 9

Integration Notes

Project: Human-Computer Interaction (HCI) Clinical Decision Support System

1. System Inputs

Manual Entry

- Patient demographics (age, sex, patient ID)
- Chief complaint
- Pain score
- Arrival mode
- Nurse observations and clinical notes

Medical Device Inputs

- Heart rate
- Blood pressure
- Respiratory rate
- Oxygen saturation (SpO₂)
- Temperature

Electronic Health Record (EHR)

- Previous ED visits
- Past medical history
- Current medications
- Allergies (where available)

2. AI Processing

The CARISURG Triage Guardian prediction engine receives the available patient information and analyses it using the selected machine learning model. The model evaluates demographic characteristics, vital signs and presenting complaint to estimate the patient's triage priority.

3. System Outputs

After processing the patient information, the system displays:

- Recommended triage level
- Confidence score
- Colour-coded urgency alert
- Explanation of the factors influencing the recommendation
- Suggested clinical action (e.g., immediate physician review)
- Audit log entry recording the AI recommendation

4. Human Interaction Following AI Recommendation

The triage nurse reviews the recommendation and its explanation before making a clinical decision.

The nurse may:

- Accept the recommendation and continue the normal ED workflow; or
- Override the recommendation if additional clinical assessment indicates a different priority.

When an override occurs, the nurse records the reason, ensuring transparency and supporting future quality improvement.

5. Workflow Summary

Patient arrives → Triage nurse assesses patient → Information entered into CARISURG AI → AI analyses inputs → Recommendation, confidence and explanation displayed → Nurse accepts or overrides recommendation → Final triage decision recorded in audit trail.

6. Integration Considerations

The proposed implementation integrates into the existing emergency department workflow without replacing clinician judgement. The system is intended to complement current practice by providing explainable decision support, maintaining human oversight and creating an auditable record of AI-assisted triage decisions.